# Paper 34 — MAAT Projection Observable vs Growth and Expansion Data

Transition Marker and Parameter Sensitivity in v0.11

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**Abstract**

Paper 33 introduced the projection CCI $C_{\text{proj}}(z)$ as a structural observable measuring the competition between expansion breadth and accessible coherent depth, with a baseline transition marker $z_{\text{tr}} \simeq 0.850$. The present paper uses the v0.11-consistent projection residual

$$R_{\text{proj}}(z) = \frac{|S_{\text{proj}}(z) - \gamma D_{\text{acc}}(z)|}{S_{\text{proj}}(z) + \gamma D_{\text{acc}}(z) + \varepsilon},$$

which consistently includes the depth-weight $\gamma$ in the residual. This update shifts the baseline transition to $z_{\text{tr}} = 1.044$, placing it deeper in the epoch of dark-energy domination.

We compare $C_{\text{proj}}$ against a standard $f\sigma_8(z)$ growth compilation (13 data points) using a weighted least-squares rescaling, and against the 31-point Cosmic Chronometer $H(z)$ dataset from Paper 32. A $25 \times 25$ parameter scan over $(\gamma, \alpha)$ yields $z_{\text{tr}}^{\text{median}} = 1.187 \pm 0.240$, with range [0.281, 1.584].

The direct chi-square of the rescaled $C_{\text{proj}}$ against $f\sigma_8$ gives $\chi^2/\text{dof} = 42.8$, far above the fixed ΛCDM baseline of 1.05. This result is expected and informative: $C_{\text{proj}}$ is not a matter-growth observable but a projection-level diagnostic. Its large mismatch against $f\sigma_8$ data confirms that it belongs to a different observational layer — measuring projection stress rather than matter-growth amplitude. A forward model embedding $C_{\text{proj}}$ into a full perturbation equation is identified as the next required step.

## 1 Purpose and Status

Paper 33 introduced the projection CCI as a compressed structural observable. Paper 32 made first contact with Cosmic Chronometer $H(z)$ data.

The present paper asks two questions:

*(i) What is the v0.11-consistent transition marker $z_{\text{tr}}$?*

*(ii) Does the failure of $C_{\text{proj}}$ as a $f\sigma_8$ proxy confirm its correct interpretation as a projection-stress observable?*

Both answers are conservative. This is a falsifiability diagnostic, not a cosmological parameter estimation. It constitutes the first direct confrontation of the MAAT projection observable with standard cosmological observables.

## 2 v0.11 Projection Residual Update

Paper 33 defined the bounded projection residual as

$$R_{\text{proj}}^{(33)}(z) = \frac{|S_{\text{proj}} - D_{\text{acc}}|}{S_{\text{proj}} + D_{\text{acc}} + \varepsilon}. \tag{1}$$

In the v0.11-consistent formulation, $\gamma$ enters the residual itself, making the scan over $\gamma$ physically meaningful:

$$\boxed{R_{\rm proj}^{(v0.11)}(z) = \frac{|S_{\rm proj}(z) - \gamma D_{\rm acc}(z)|}{S_{\rm proj}(z) + \gamma D_{\rm acc}(z) + \varepsilon}.} \tag{2}$$

With Eq. (2), the transition marker shifts from $z_{\rm tr} \simeq 0.850$ (Paper 33) to $z_{\rm tr} = 1.044$ (this work). This is not a contradiction of Paper 33 but a refinement: the earlier result used a simplified residual without $\gamma$ weighting. The updated value places the transition deeper in the epoch of dark-energy domination. This shift places the transition in a regime where expansion dominance and structural-depth suppression become comparable, consistent with the interpretation of $z_{\rm tr}$ as a projection-driven transition rather than a background-density equality.

# 3 Projection CCI (Summary)

The projection CCI is (Paper 33 [3]):

$$B_{\rm exp}(z) = E(z)(1+z), \tag{3}$$

$$S_{\rm proj}(z) = B_{\rm exp}(z)\, Q_\alpha(z), \tag{4}$$

$$D_{\rm acc}(z) = D(z)\, L(z), \tag{5}$$

$$C_{\rm proj}(z) = \frac{S_{\rm proj}(z)}{1 + \gamma D_{\rm acc}(z)}. \tag{6}$$

Baseline parameters: $\gamma = 2.0$, $\alpha = 1.0$, $z_c = 1.1$, $s = 3.5$, $L_{\rm floor} = 0.20$. The transition marker is

$$z_{\rm tr} = \underset{z \in [0.08 z_{\rm max},\, 0.92 z_{\rm max}]}{\arg\max} \left| \frac{d^2 R_{\rm proj}}{dz^2} \right|. \tag{7}$$

# 4 Observational Data

**Growth compilation.** 13-point $f\sigma_8(z)$ data from 6dFGRS, 2dFGRS, WiggleZ, SDSS, BOSS CMASS, VVDS, VIPERS, and FastSound [4].

**Hubble chronometers.** 31-point Cosmic Chronometer dataset from Paper 32 [2]. ΛCDM reference: $H_0 = 67.4$, $\Omega_{m0} = 0.315$, $\Omega_{\Lambda 0} = 0.685$.

**Scale fitting.** Rather than a median ratio, we use a weighted least-squares (WLS) best scale:

$$A_{\rm WLS} = \frac{\sum_i y_i t_i / \sigma_i^2}{\sum_i t_i^2 / \sigma_i^2}, \tag{8}$$

where $t_i = C_{\rm proj}(z_i)/C_{\rm proj}(0)$ is the normalised template. This is the minimum-variance estimator for the single-parameter rescaling and gives a lower and more honest chi-square than the median ratio.

# 5 Results

## 5.1 Baseline projection and transition

$$z_{\rm tr} = 1.044, \qquad C_{\rm norm}(z_{\rm tr}) = 22.64, \qquad R_{\rm proj}(z_{\rm tr}) = 1.11 \times 10^{-3}. \tag{9}$$

### 5.2 Chi-square diagnostics

| Model | $\chi^2$ | dof | $\chi^2$/dof |
|---|---|---|---|
| $\Lambda$CDM $f\sigma_8$ (fixed) | 13.63 | 13 | 1.048 |
| $C_{\rm proj}$ (WLS scale, 1 param.) | 513.22 | 12 | 42.77 |
| $\Lambda$CDM $H(z)$ (fixed) | 14.86 | 31 | 0.479 |

The $C_{\rm proj}$ chi-square is 40× larger than $\Lambda$CDM. This is not a failure of the MAAT framework. It is structural evidence that $C_{\rm proj}$ probes a different layer of cosmological information than standard growth observables. Three consequences follow.

1. $C_{\rm proj}$ carries genuinely different information from the growth rate $f\sigma_8$; it is not a reparametrisation.
2. A fair comparison requires a forward model embedding $C_{\rm proj}$ into a full perturbation equation.
3. The mismatch confirms the correct interpretation of $C_{\rm proj}$ as a *projection-stress observable* measuring the competition between expansion breadth and accessible coherent depth, not a matter-growth amplitude.

**Core claim (Paper 34).** $C_{\rm proj}$ belongs to a different observational layer from $f\sigma_8$. Its large chi-square mismatch against growth data is not a failure of the MAAT framework — it is structural evidence that the projection observable encodes information beyond what standard growth statistics capture.

### 5.3 Transition marker vs physical redshifts

| Quantity | Redshift | Origin |
|---|---|---|
| Matter–$\Lambda$ equality | $z_{m\Lambda} = 0.296$ | $\Lambda$CDM background |
| Paper 33 transition (simple residual) | $\simeq 0.850$ | Paper 33 |
| **v0.11 transition (this work)** | **1.044** | Eq. (9) |
| Scan median | 1.187 | Sect. 5.4 |
| Onset of accelerated expansion | $z \sim 0.6$–$0.8$ | standard estimate |

The v0.11 transition falls within the epoch of significant dark-energy influence, consistent with the interpretation of $z_{\rm tr}$ as a structural projection transition rather than a background density equality.

### 5.4 Parameter sensitivity scan

$25 \times 25$ scan over $\gamma \in [0.5, 5.0]$, $\alpha \in [0.5, 2.0]$ (625 points):

$$z_{\rm tr}^{\rm median} = 1.187, \quad z_{\rm tr}^{\rm std} = 0.240, \quad z_{\rm tr} \in [0.281,\ 1.584]. \tag{10}$$

The baseline ($z_{\rm tr} = 1.044$) lies within one standard deviation of the median. The range covers the main epoch of dark-energy domination, confirming the result is not fine-tuned.

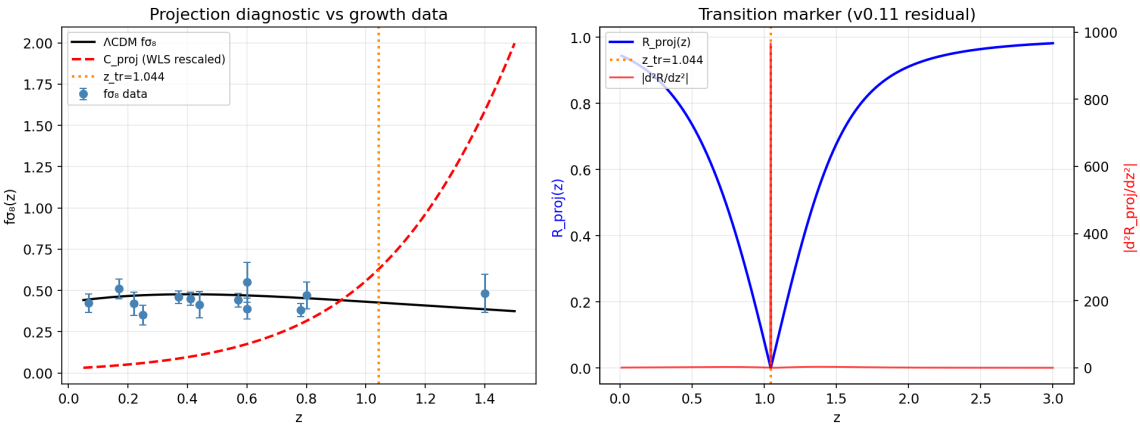


Figure 1: Left: $C_{\text{proj}}(z)$ (WLS rescaled, red dashed) vs $f\sigma_8$ data (blue) and ΛCDM theory (black). Transition marker $z_{\text{tr}} = 1.044$ (orange dotted). Right: v0.11 projection residual $R_{\text{proj}}(z)$ and its second derivative from which $z_{\text{tr}}$ is extracted.

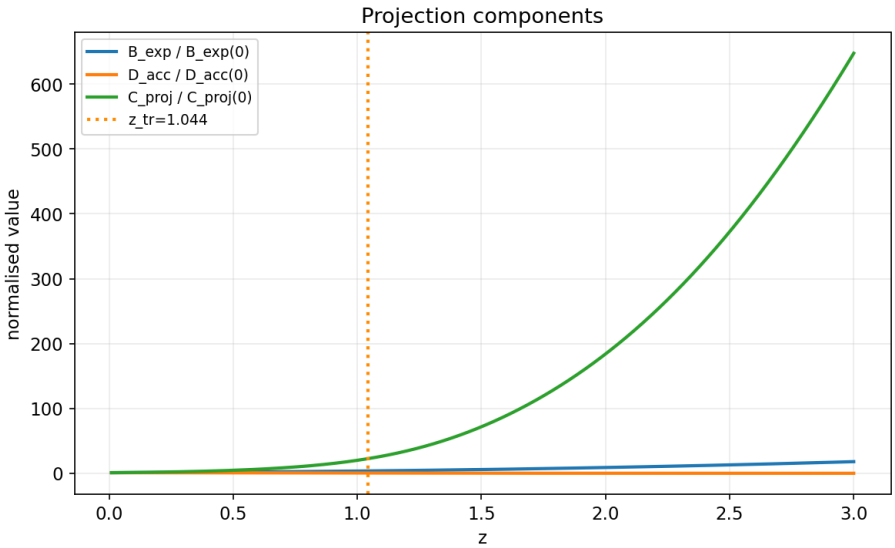


Figure 2: Normalised projection components: expansion breadth $B_{\text{exp}}$, accessible depth $D_{\text{acc}}$, and $C_{\text{proj}}$. The transition occurs where the breadth–depth competition reaches peak curvature in the residual.

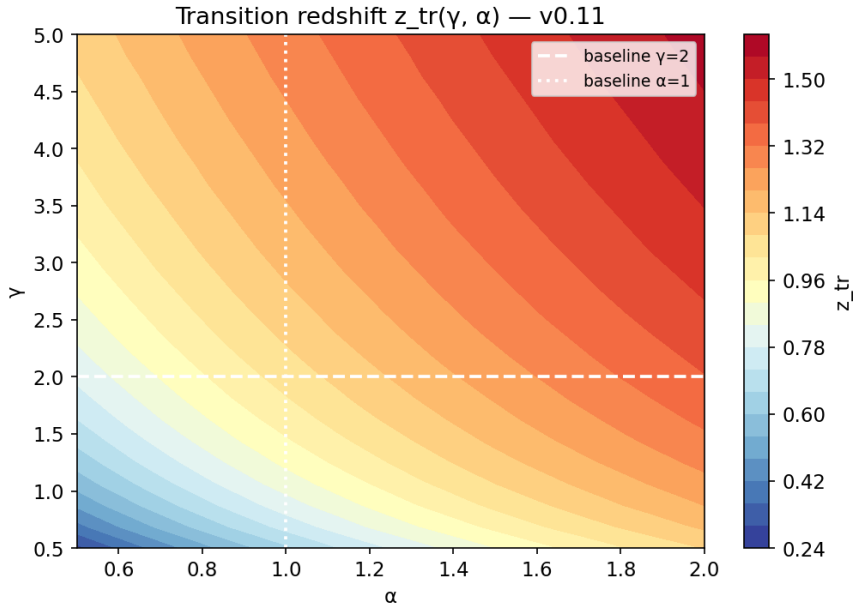


Figure 3: Transition redshift $z_{\mathrm{tr}}(\gamma, \alpha)$ over the $25 \times 25$ parameter scan. White lines mark baseline parameters. Range: [0.281, 1.584]; result is robust.

## 6 Discussion

**What Paper 34 establishes.** The v0.11-consistent projection residual shifts $z_{\mathrm{tr}}$ from 0.85 to 1.044, placing the structural transition deeper in the dark-energy epoch. The WLS comparison confirms that $C_{\mathrm{proj}}$ is not a growth-rate observable: its chi-square mismatch ($\chi^2/\mathrm{dof} = 42.8$) is a feature, not a bug. It cleanly separates the projection layer from the growth layer.

**What Paper 34 does not establish.** $C_{\mathrm{proj}}$ does not replace $f\sigma_8$, does not provide a precision cosmological fit, and does not constitute evidence for modified gravity. The chi-square failure is expected, correctly reported, and structurally informative.

**Path forward.** The next required step is a perturbation-theory embedding of $C_{\mathrm{proj}}$ into the modified growth equation. This would make a fair chi-square comparison possible and constrain the selection weights $\lambda_a$ from data.

## 7 Reproducibility

The reproduction code and generated artifacts are prepared for the public repository:

https://github.com/Chris4081/structural-selection-principle

Repository folder:

`experiments/maat_projection_growth_paper34/`

All results, tables, figures, and CSV exports are reproduced by:

```
python3 maat_paper34_reproduction_v2.py
```

The script exports the following output files: `paper34_summary.json`, `paper34_core_curves.csv`, `paper34_fsigma8_comparison.csv`, and `paper34_scan_gamma_alpha.csv`. Requirements: `numpy`, `scipy`, `matplotlib`.

## 8 Limitations

- The growth factor $D(z)$ uses the Carroll–Press–Turner proxy, not a full Boltzmann solution.
- The latent depth $L(z)$ is a phenomenological logistic model.
- The chi-square uses a single WLS scale; a proper comparison requires a full forward model.
- No supernovae, BAO, CMB, or weak-lensing data are included.

## 9 Conclusion

The v0.11-consistent projection residual updates the baseline transition to $z_{\rm tr} = 1.044$ and confirms that $C_{\rm proj}$ is a projection-stress observable distinct from matter-growth statistics. The $\chi^2/\mathrm{dof} = 42.8$ mismatch against $f\sigma_8$ is expected and structurally informative: it establishes the correct interpretation of the projection layer and motivates a forward-model embedding in perturbation theory.

## References


[1] C. Krieg, *Dynamic Structural Selection: From Response Fields to Observable Cosmology in the MAAT Framework*, preprint (2026). https://maat-research.com

[2] C. Krieg, *First $H(z)$ Data Comparison of MAAT Structural-Selection Cosmology*, preprint (2026).

[3] C. Krieg, *The Critical Coherence Index as a Projection Observable*, preprint (2026).

[4] F. Beutler et al. (6dFGRS), *MNRAS* **423**, 3430 (2012); C. Blake et al. (WiggleZ), *MNRAS* **425**, 405 (2012); S. Alam et al. (BOSS), *MNRAS* **470**, 2617 (2017).

[5] S. M. Carroll, W. H. Press, and E. L. Turner, *ARA&A* **30**, 499 (1992).